

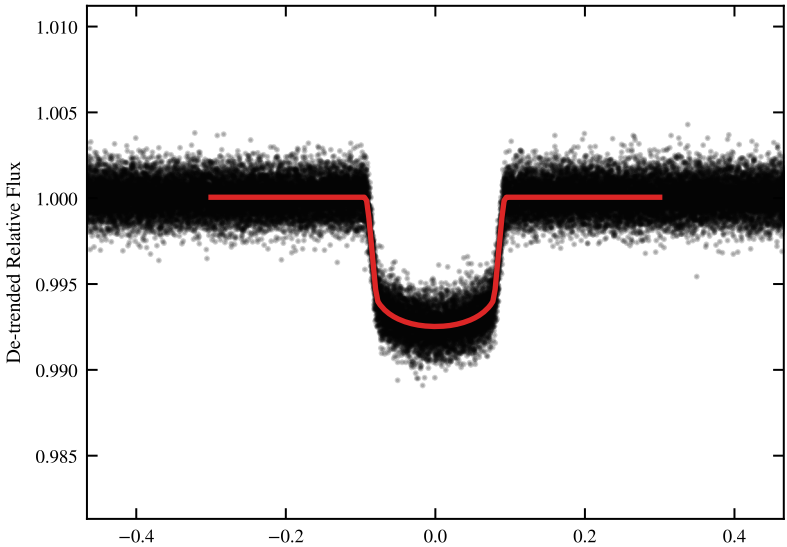
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 287196418 | RA: 292.73413° | Dec: 59.40385° | Sectors: 53, 54, 55, 56, 57, 58, 59, 60

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	3.695 ± 4.3e-06
Rp/R★	0.08233 ± 0.000265
Rp (R⊕)	14.686 ± 0.421
b	0.28141 ± 0.0434
T14 (h)	4.4732 ± 0.00897
a (AU)	0.050374 ± 0.00154
Teq (K)	1917.1 ± 31.8

Primary Phased Transit



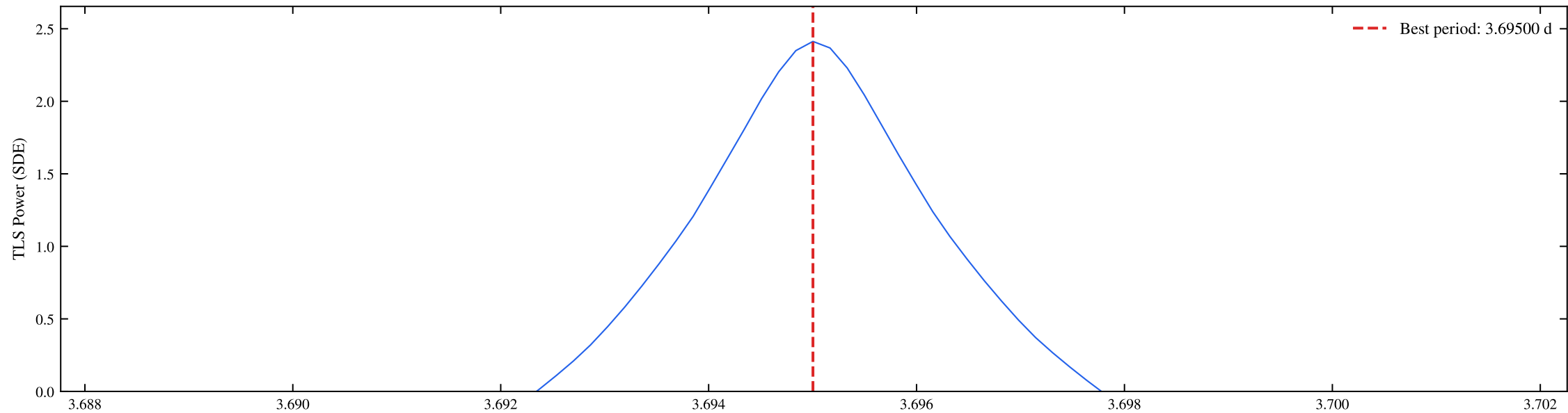
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	3.695	3.695015 ± 0.00000
P1 Epoch (BTJD)	N/A	2744.6776 ± 0.0001
Max R-hat (R)	N/A	1.0008
Min ESS	N/A	7144
MCMC Divergences	N/A	0

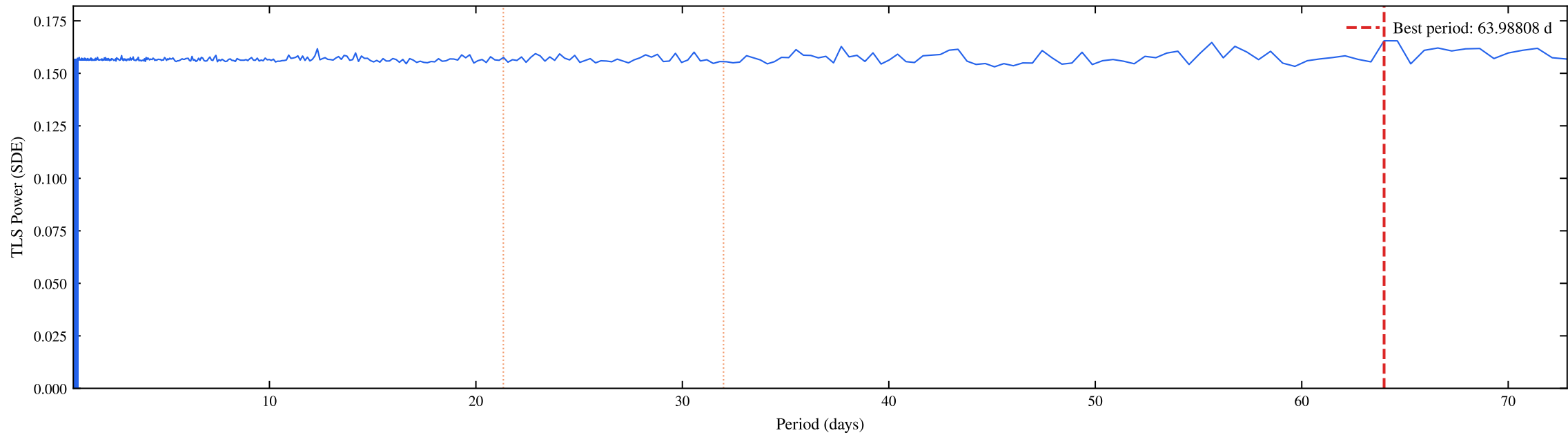
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.634 ± 0.046	VizieR TIC8
M★ (M☉)	1.532 ± 0.186	VizieR TIC8
Teff (K)	6982 ± 108	VizieR TIC8
log g (dex)	4.197 ± 0.0916	VizieR TIC8
[Fe/H]	-0.524 ± 0.0157	Gaia DR3
ρ★ (g/cm³)	0.4951 ± 0.0732	Derived

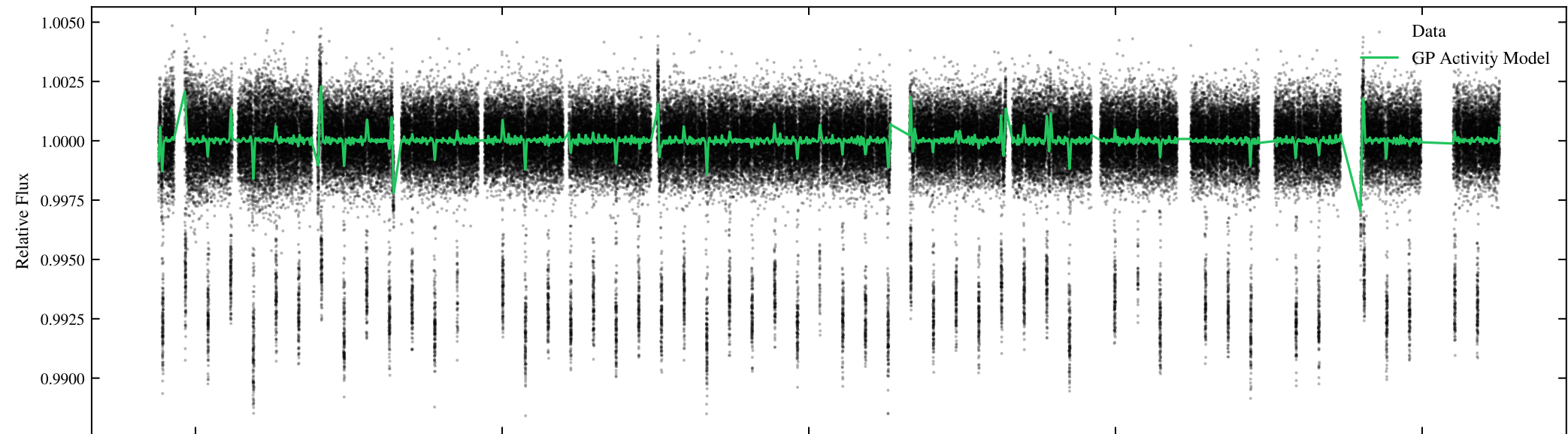
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Periodogram (Planet 1 Search) (SDE = 2.41)



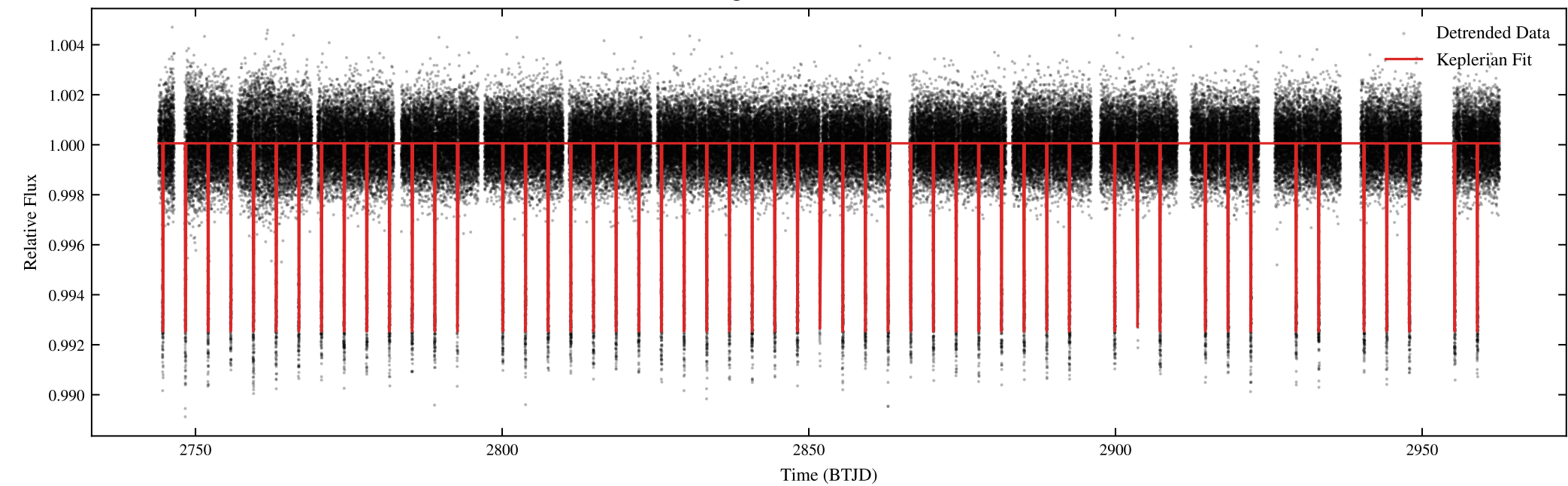
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Periodogram (Planet 2 Search) (SDE = 0.17)



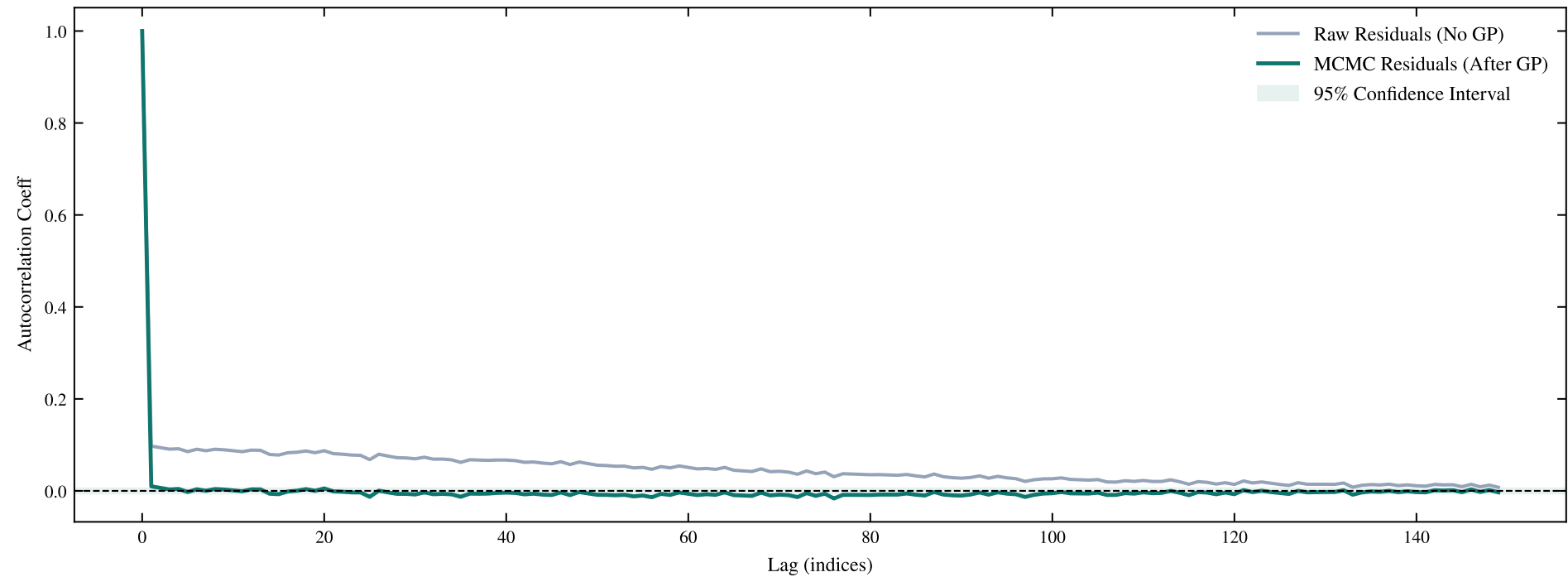
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | Bayesian GP Detrending (Time Domain)



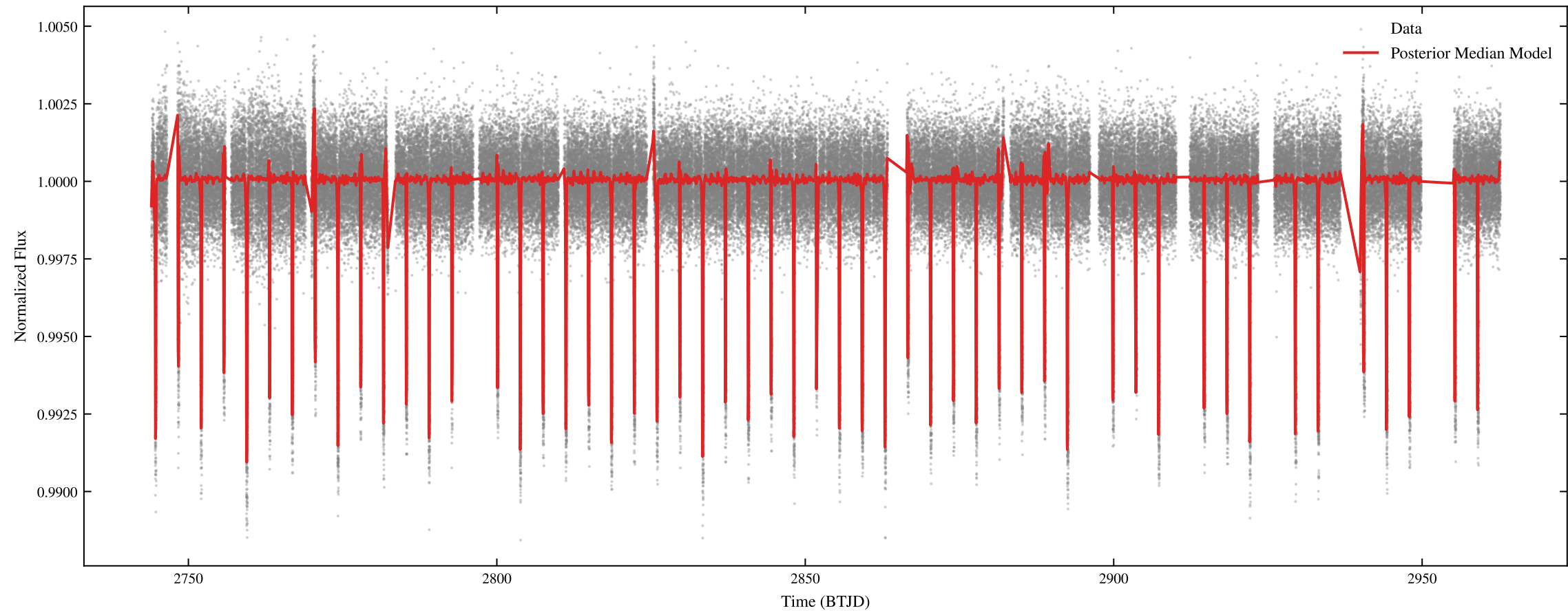
Detrended Light Curve & Best-Fit Transit Model

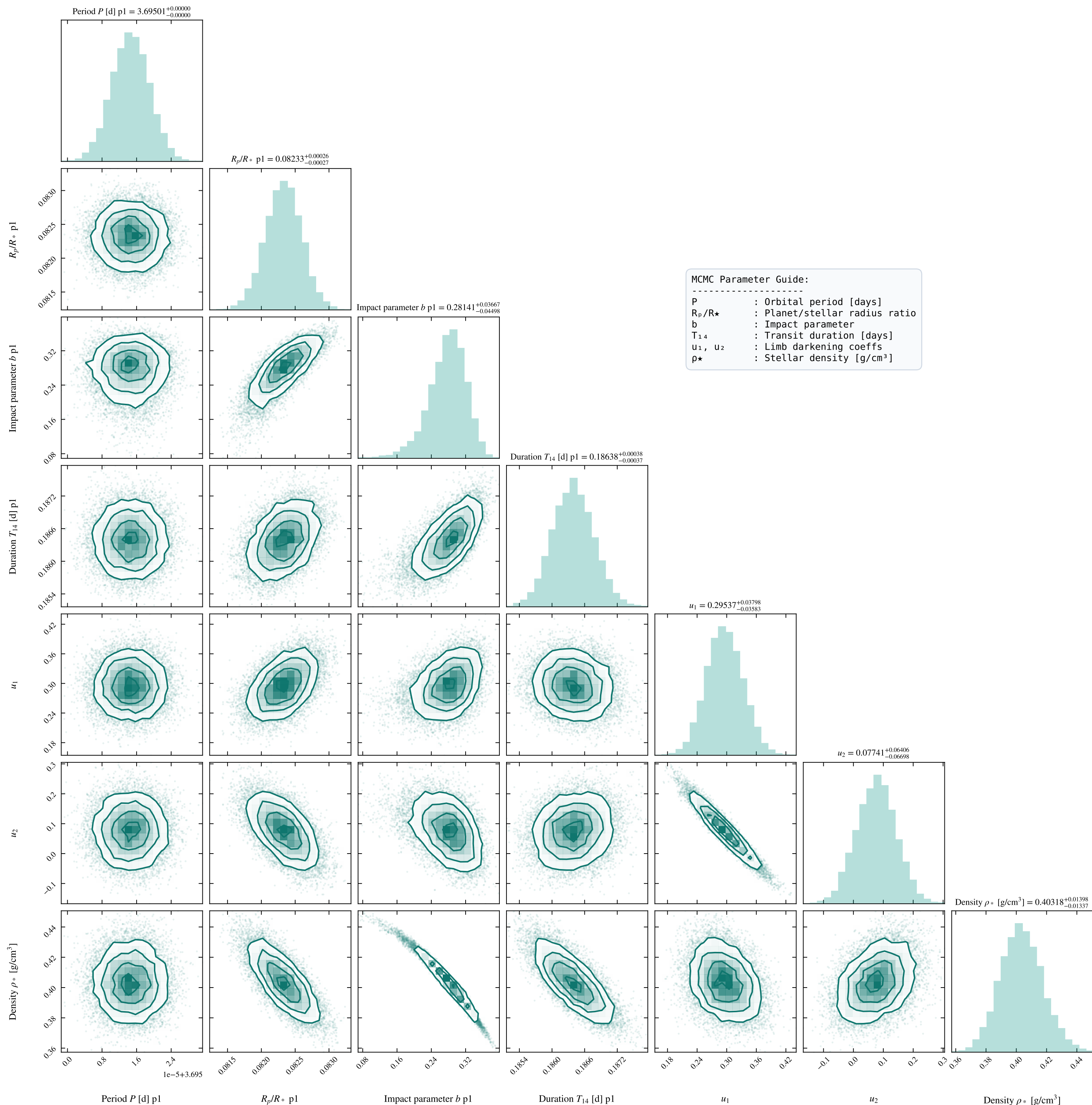


TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | GP Residuals Autocorrelation Function (ACF)

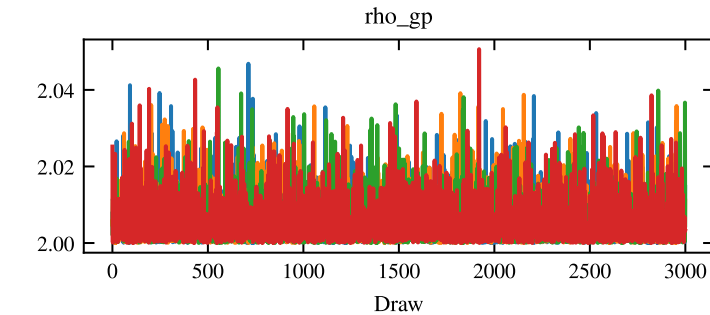
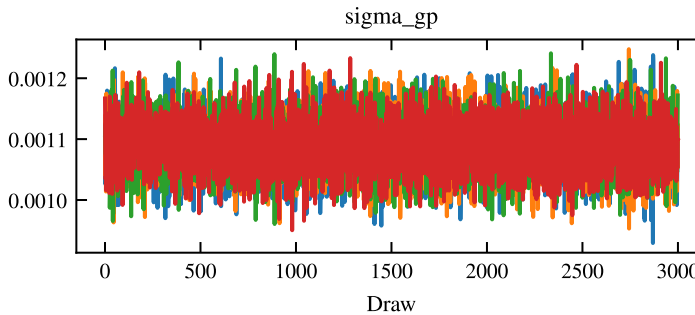
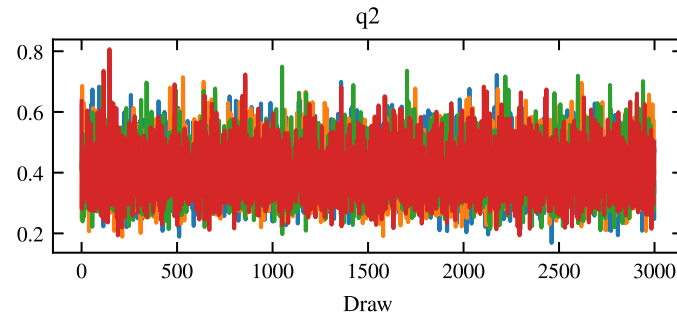
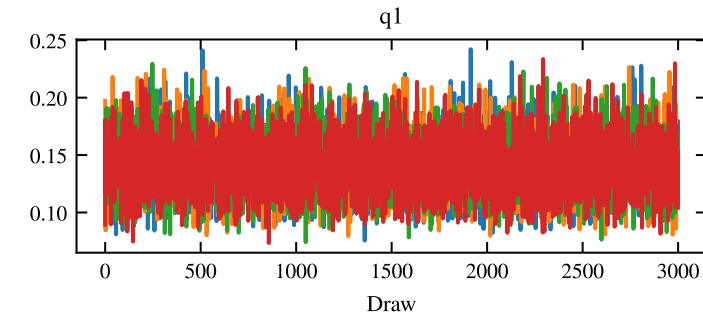
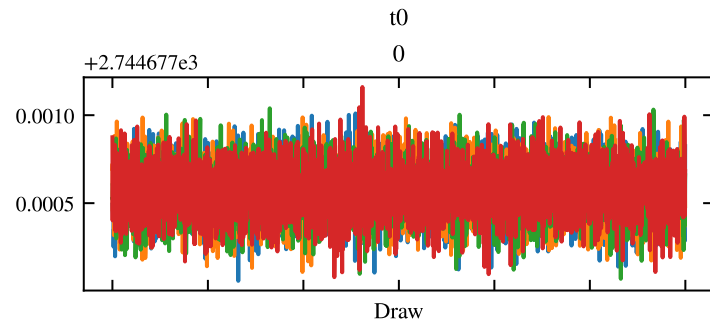
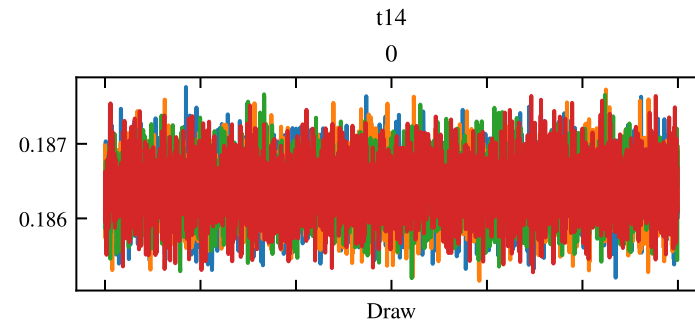
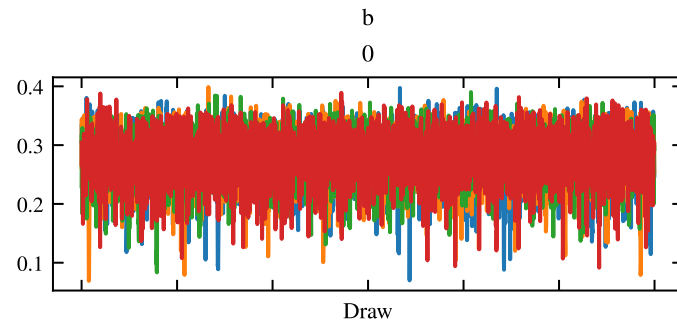
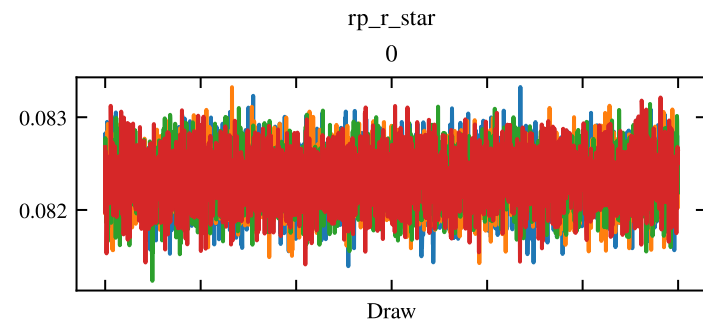


TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | MCMC Posterior Predictive Check

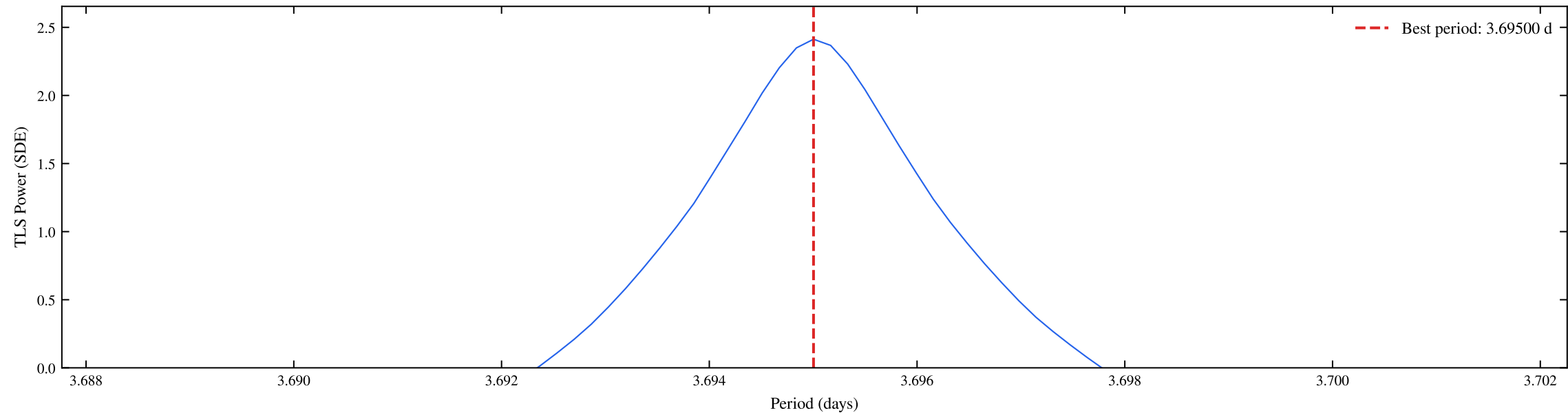




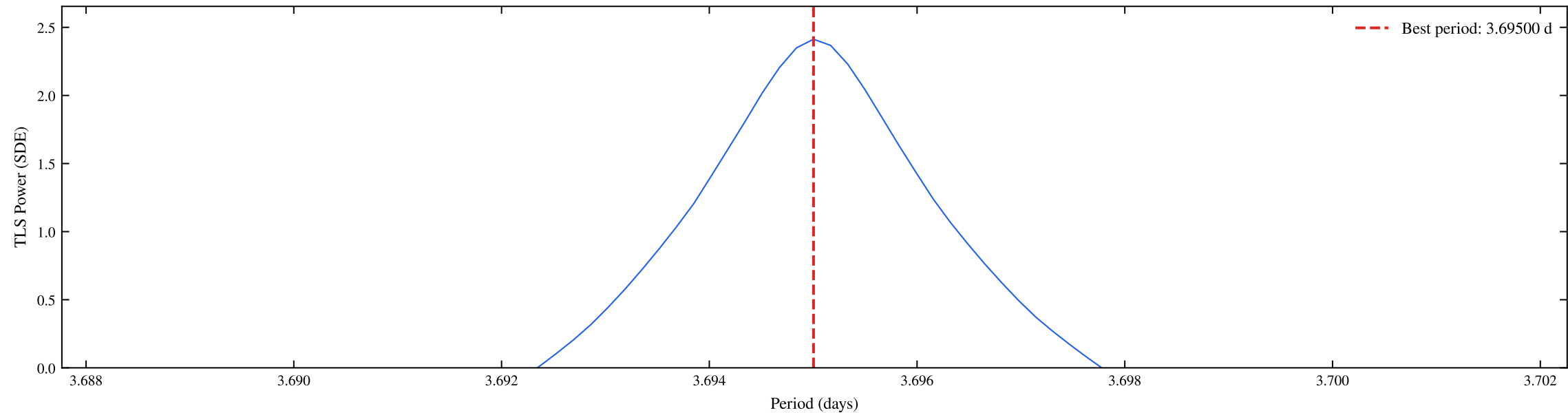
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | MCMC Trace Plots



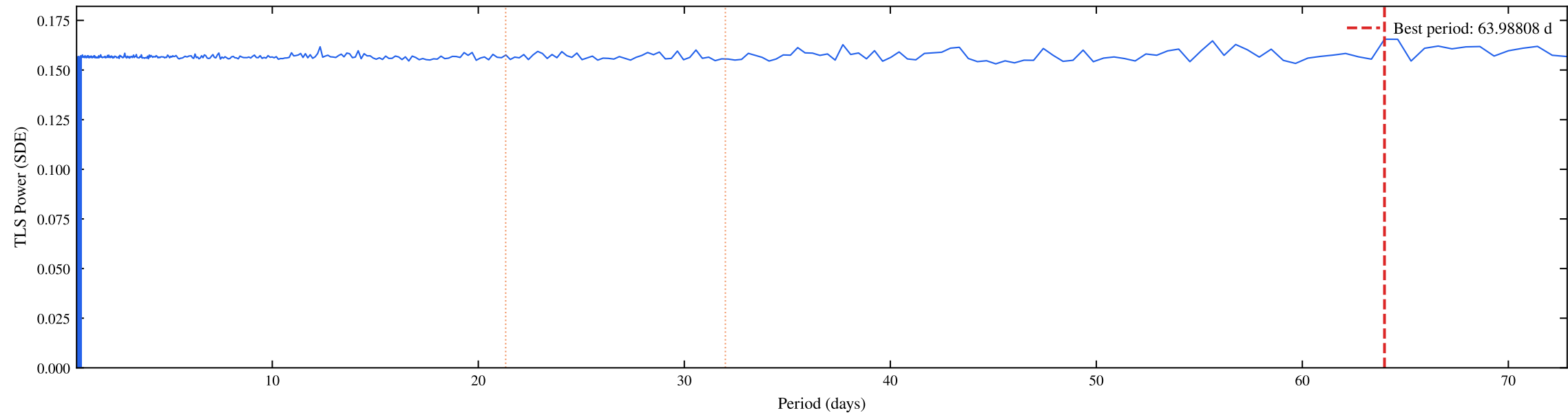
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Coarse Periodogram (SDE = 2.41)



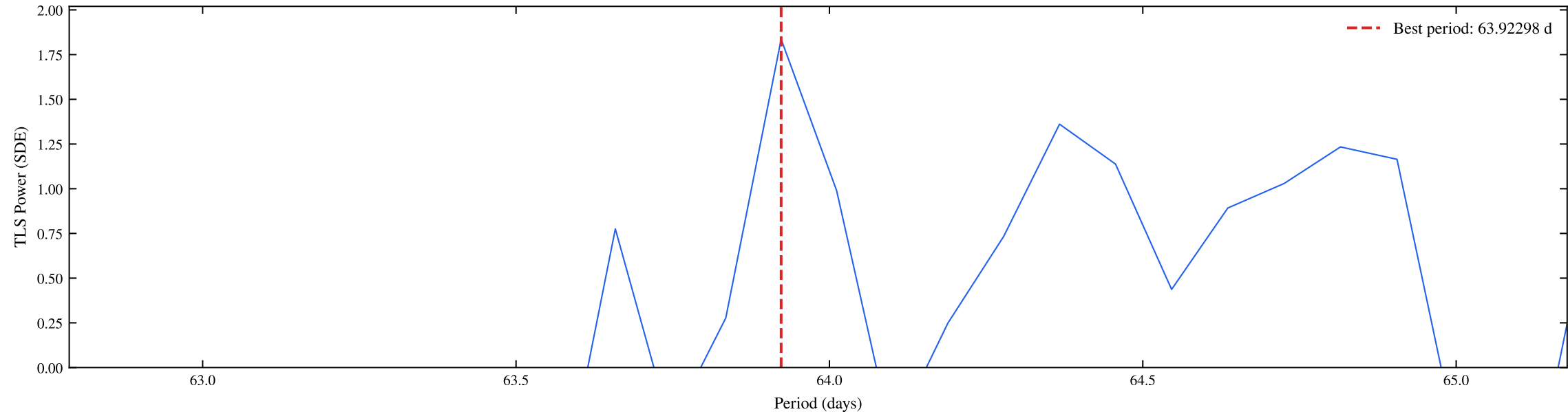
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Fine Periodogram (SDE = 2.41)



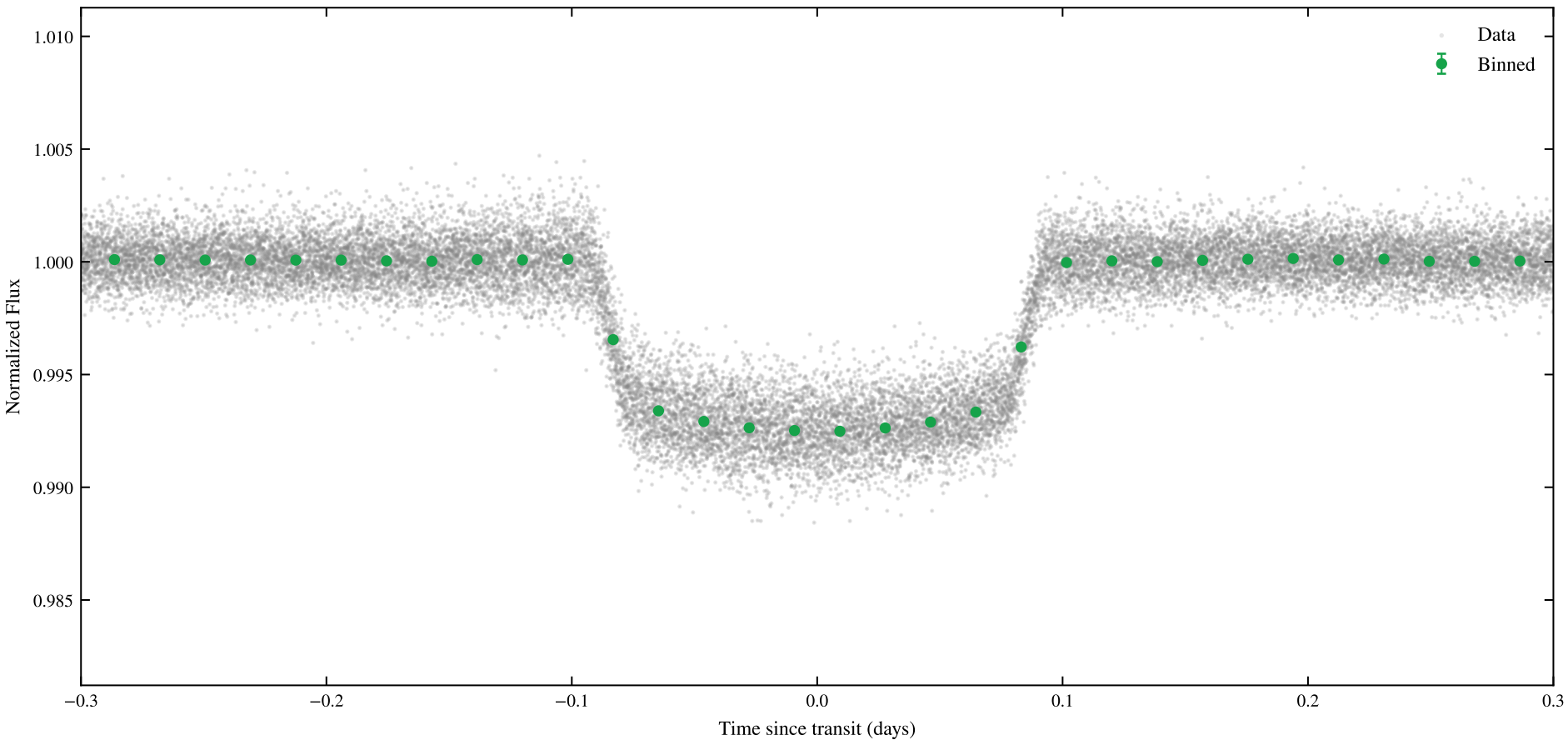
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Coarse Periodogram (SDE = 0.17)



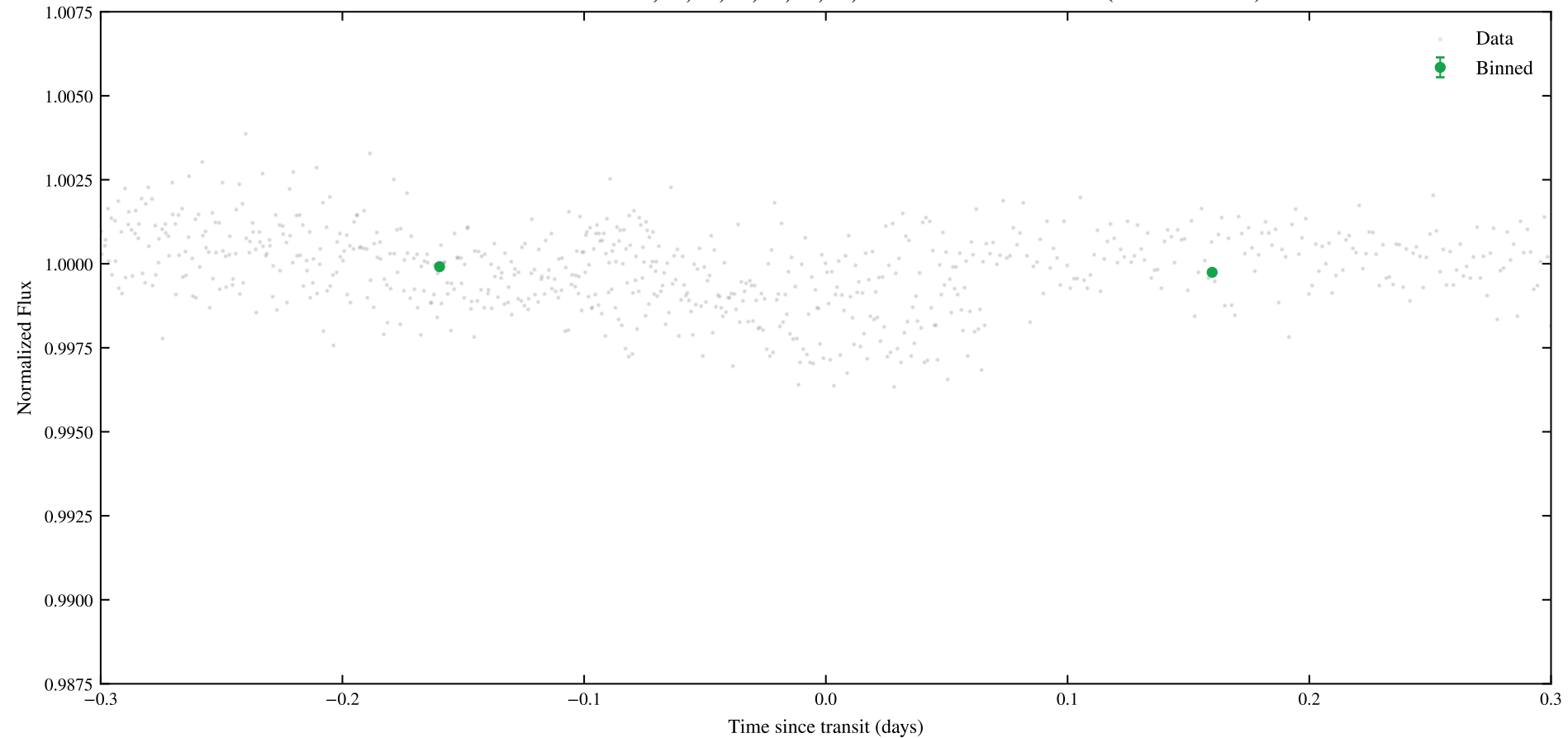
TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | TLS Fine Periodogram (SDE = 1.84)

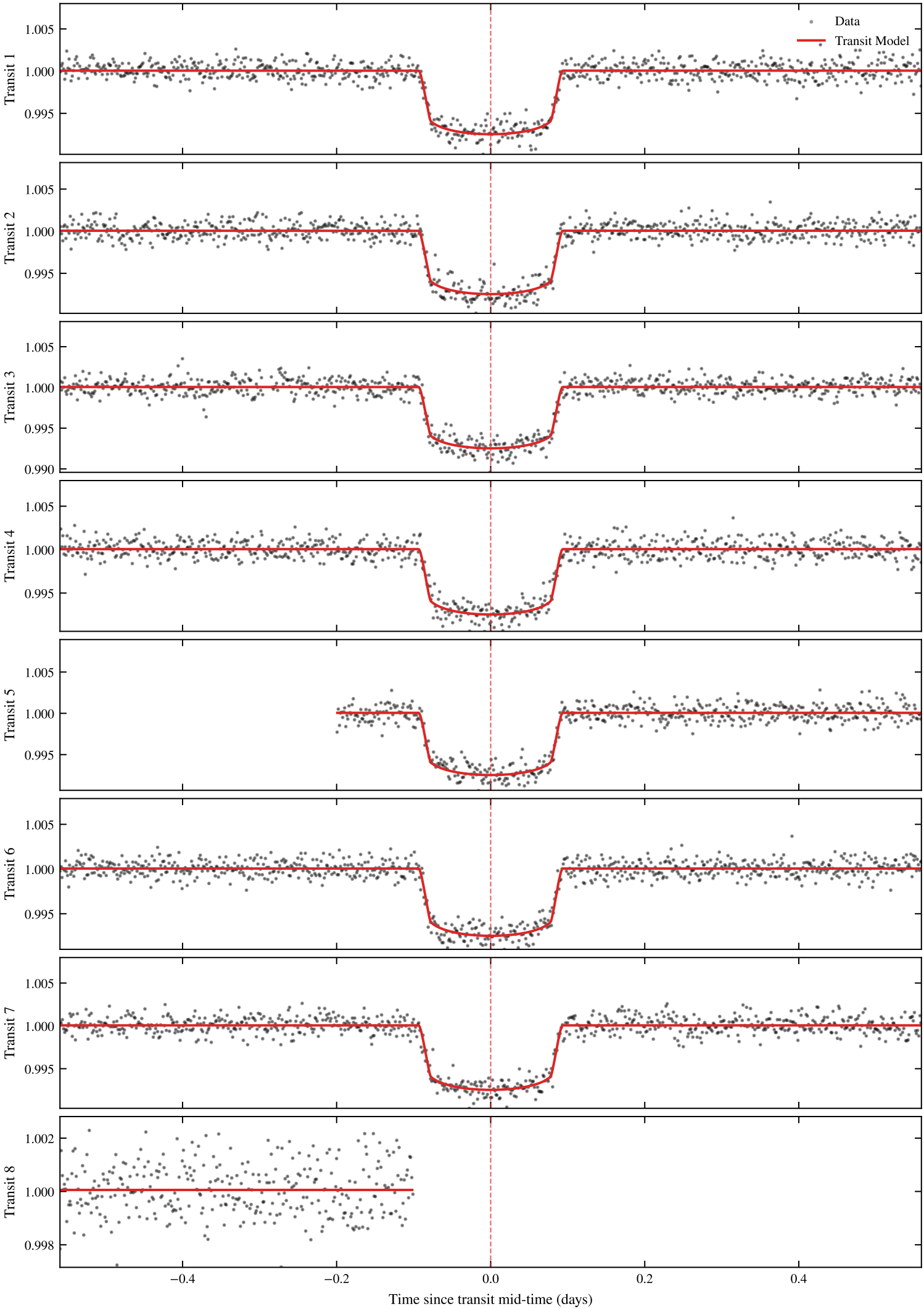


TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | Phase-Folded Transit (P = 3.69500 d)



TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | Phase-Folded Transit (P = 63.92298 d)





TIC 287196418 | Sectors: 53, 54, 55, 56, 57, 58, 59, 60 | Phased MCMC Fit - Planet 1

